

EMMANUEL MDLULI

Faculty of Computer Science · Co-op Program

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WORK EXPERIENCE

Software Development Intern— Steed Motors Inc NB

June 2026 – Present · Fredericton, NB

- Engaged on a paid multi-month retainer to design, develop, and maintain bespoke software for a dealership — data visualizations, internal automation tools, and website maintenance.
- Integrated agentic AI tooling (Claude Code, IDE-embedded assistants, LLM-driven code review) end-to-end across design, implementation, debugging, and refactoring — compressing build and deployment timelines.
- Shipped automated site-testing scripts (Python, Playwright); now building a full-stack inventory advertising and tracking tool.

PERSONAL PROJECTS

Engineering Drawing Review System — CoLab Inspired

Python · React · FastAPI · SQLite · Anthropic Claude API · JWT (Passlib + bcrypt, python-jose)

- Built a full-stack AI-powered CAD drawing reviewer (React, FastAPI) using Claude Vision to detect missing dimensions, geometry errors, and unclear tolerances — rendering bounding-box pins directly on the drawing.
- Designed a multi-user collaboration system — engineers join shared project accounts to upload, analyze, and track drawings, with per-run analysis history and open/resolved issue tracking.
- Implemented JWT-authenticated REST API with SQLite persistence, supporting AI and manual issue pins, critical/major/minor severity classification, and independent re-analysis runs stored for side-by-side comparison.

AWS Risk Visualizer — AWS IAM Permissions Auditor

Python · JavaScript · D3.js · AWS Boto3 · NetworkX

- Built an interactive cloud-security auditor that pulls live IAM roles and policies from AWS via Boto3 and renders a D3.js force-directed graph of identity → policy → resource relationships.
- Implemented per-statement risk classification (wildcard exposure, privilege escalation, lateral movement) with a critical / warning / info severity model surfaced in a dedicated findings panel.
- Aggregated 500+ resource ARNs per policy into per-service graph nodes with hover-revealed detail, collapsing a hairball visualization into a readable overview.

Claims Copilot — AI-Assisted Insurance Claims Processing

Python · FastAPI · PostgreSQL · Next.js · TypeScript · Anthropic Claude API · JWT · Docker

- Built a full-stack insurance claims assistant that ingests policy PDFs, classifies claims, and drafts bilingual (EN/FR) response letters with exact clause citations — every draft requiring human approval before finalization.
- Designed a retrieval-augmented drafting pipeline — policy chunks indexed and ranked in PostgreSQL feed a two-model LLM architecture that controls API cost without sacrificing draft quality.
- Implemented a JWT-authenticated REST API with a full audit trail — every retrieval, LLM call, edit, and approval recorded as structured events — enforcing a pending → drafted → approved/rejected workflow.

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Java, Kotlin, C, Python, SQL, HTML & CSS, MIPS Assembly, Racket, Octave

Tools & Technologies: Git & GitHub, AWS, MySQL, PostgreSQL, VS Code, Boto3, D3.js, Node.js, Vite, React, FastAPI, Pydantic, SQLite3, JavaFX, .NET / ASP.NET, NetworkX, NumPy, JUnit, Playwright, UML, Microsoft Office (Word, Excel, PowerPoint, Outlook)

Agentic Tooling: Claude Code, Claude API, agentic development workflows, IDE-integrated AI assistants

EDUCATION

University of New Brunswick · Fredericton, NB — Bachelor of Computer Science (September 2023 – Present) · Current Term GPA: 3.35 (Winter 2026)

AWARDS / ACADEMIC EXPERIENCE

- FreeCodeCamp HTML & CSS Certification · Coursework: information security, software engineering, databases.